



अश्ववा MOBILITY FOUNDATION

MARCH 2023 EDITION

LIVE IT. LOVE IT. RACE IT.



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EVENTS & NETWORKING ACTIVITIES

A sneak peek into the events and
team networking activities!



AEROMANIA 2023

Aeromania Day was a highly anticipated event for aviation and automotive enthusiasts and Ashwa Mobility's participation made it even more special. Ashwa Mobility was invited to demonstrate the performance of their cars, adding a new dimension to the event.

The event was held at RVCE on the 4th, 5th, and 6th of January, and a large crowd had gathered to witness the display. The Ashwa Mobility team arrived in their car model on the third day, with impressive specifications and unique features.



Demonstration of the Ashwa Formula Student Car
[Real time video hyperlinked](#)

As the engines roared to life, the spectators were treated to a thrilling display of speed and manoeuvrability. The cars darted across the roads of RVCE, demonstrating their impressive acceleration and handling capabilities. Overall, Ashwa Mobility's participation in Aeromania Day was a resounding success. The event provided a platform for showcasing the impressive performance of their cars, leaving a lasting impression on all those who attended.



The Ashwa team at the Aeromania event



IMTEX - 2023

IMTEX 2023 was organised by Indian Machine Tool Manufacturers' Association (IMTMA) and was held from 19th to 25th January at Bangalore International Exhibition Centre. There were about 991 exhibitors, with more than 20+ countries participating.



The various exhibits and booths at the IMTEX Event

Divisions :

- TOOLTECH - showcased machine tool accessories, metrology and CAD/CAM cutting tools, tooling systems and trends
- Digital Manufacturing - showcased the latest products in the Additive Manufacturing and Industry 4.0
- Institute-industry ACADEMIA PAVILION - Academic / R&D Institutions showcased their R&D capabilities for the betterment in the marketplace

We interacted with companies in the automobile industry and found out more about trending technology. Also had the chance to meet with industrial experts with respect to the technology related to mobility. Overall, it was a great experience.



OPEN DAY AT THE ASHWA WORKSHOP

The Open Day of Ashwa Mobility, held on 2nd February, 2023 was a highly anticipated event for both the students and staff. The event was made even more special as the principal of the institution paid a visit to witness the proceedings firsthand. As soon as the workshop was opened, students and visitors poured in to witness the latest advancements in mobility technology. The event featured several displays and demonstrations of performance vehicles and innovative mobility solutions.



Interaction with the Principal and other faculty at the Open Day

The principal toured the exhibits, taking an active interest in the various vehicles and technologies on display. He interacted with the students, asking insightful questions and providing valuable feedback. Overall, the Open Day was a resounding success, with the principal commending the students and staff for their efforts. The event served as a platform for showcasing the institution's commitment to innovation and sustainability in the field of mobility, leaving a lasting impression on all those who attended.



A Group shot along with our faculty coordinator,
Dr. Ravindra S Kulkarni and other dignitaries of the college.



RECRUITMENT DRIVE

Ashwa is always expanding! Many eager students applied for the most recent round of recruiting that started on the 6th of February.

An aptitude exam and an interview round were both conducted as part of the hiring process. The aptitude test measures how they can solve analytical problems and the interview procedure reveals their motivation and capacity for learning.

We noticed that an increasing number of students share Ashwa's passion and zeal. Everyone that was involved found the experience to be fruitful!



Keen students taking the Ashwa aptitude test



First interaction with the 2026 batch of students post recruitment

PROJECT FOCUS

FORMULA STUDENT

Developments and future
goals



FORMULA HYBRID: KEY GOALS AND DEVELOPMENTS

Ashwa's Hybrid Formula Student prototype RZ-X5H was first developed in the year 2015. The Series Hybrid prototype thus developed since 2015 was aimed to operate at maximum efficiency and further advancement lead to the design and fabrication of the parallel hybrid architecture in 2021. The Parallel Hybrid thus developed was showcased in the competition and won the first place second time in a row at the Formula Hybrid USA Event overall globally bagging 3 top awards.

The goal since 2020 has been to build a parallel hybrid powertrain which can have a 10% increase in efficiency than the previous series powertrain prototypes which was accomplished in 2021 successfully.



Ashwa Formula Hybrid Prototype

The reasons for choosing parallel architecture are twofold over that of series:

- **Performance:** A Parallel Hybrid Architecture allows us to switch the power source in the car between the engine and motor dynamically for different dynamic events and allows us to extract maximum possible performance in different conditions.
- **Efficiency:** Our parallel hybrid is based on power coupling using a set of overrunning

clutches which yields high efficiency. The ability of the engine to directly drive the car instead of recharging the batteries, gives us very low power losses in energy conversion. Further, the decreased dependence on critical electrical components that are prone to faults, makes the car more reliable overall.



Ashwa's Parallel Hybrid architecture got a patent filed successfully on 19th Jan, 2022 marking yet another milestone towards its contribution in the field of greener mobility aligning with the technology as a student team.



Ashwa Batch 2022 post triumphs at USA Virtual Event for the Formula Hybrid Category.
Group shot with our faculty coordinator Dr. Ravindra S Kulkarni (left) and Principal, Dr. K N Subramanya (Right)



DYNO TUNING AT RACECONCEPTS

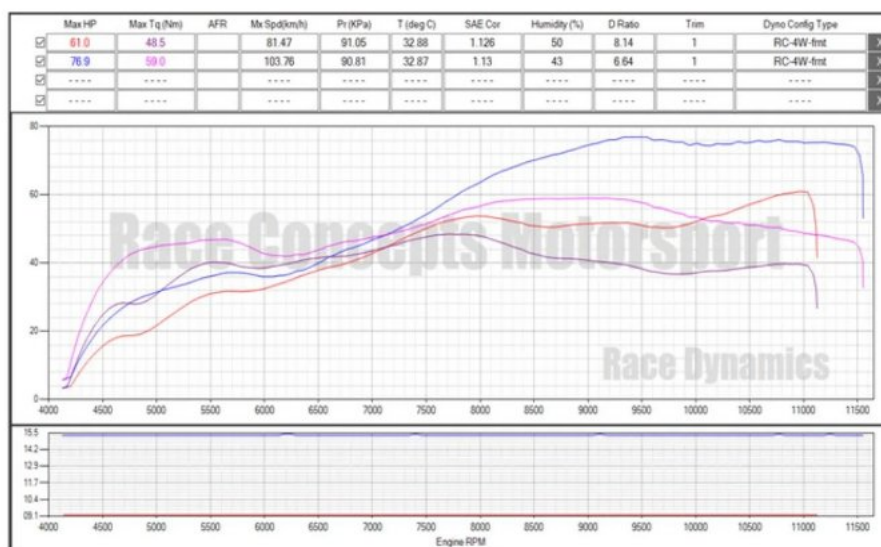


The Formula Student team tuned the Motec M800 ECU of the Combustion prototype at Race Concepts, Electronic City, Bangalore on 16th November, 2022.



The real-time video of car tuning is embedded & hyperlinked.

A new engine developed was tuned at Race Concepts using the Dynamometer. The car proved to achieve the maximum brake horse power of 79 BHP with 60 Nm peak torque upon tuning. Hence, the key goals of our engine to be more powerful, drivable, and efficient was accomplished leading to an high performance engine, dynamic handling characteristics with providing better fuel economy and smoother response. All in all, the vehicle's overall performance was improved and optimized.



Dyno report indicating the major improvements on the Linear power delivery and good mid-range torque.



TESTING AT ARUANI GRID



The Formula Student team tested the Combustion prototype at the Aruani Grid Race track on the 3rd and 4th of January 2023.



The tech senior and junior team at the Aruani grid.

The real-time video of the actual race at the test track has been embedded & hyperlinked.

Aruani grid is a 1 kilometer track with quite a few tricky turns and a 150 meter straight where top speed can be achieved. The best lap time of 56.102 seconds and acceleration run time of 4.9 seconds for 75 meters was achieved.

Different dynamic events like brake test, acceleration test and endurance run were carried out over the two day testing session.

Getting the car tested after renovation of the designs and fabrication of the prototype was a switchback journey. Although the team encountered issues at the test track, it was a fruitful experience to resolve them on-spot by digging out the root causes of the problems and concluding with remarkable progress.

Overall, it was a great outcome based learning experience at the Aruani Grid test track, we continue to make progress with numerous testing rounds at the RVCE campus as well.

TECH PILLARS

Meet the Formula Student
technical team leads of the
2024 batch

TEAM CAPTAIN

Vikhyathraj Shetty K



From a very tender age, cars and racing was what fascinated me and opting for RVCE exposed me to the most prestigious racing club, Team Ashwa Racing. Without any hesitation, I joined the team and learnt in abundance about how a car is made from the ground-up, and more importantly, how to be a team player. Working on the car gave me deep insights into the internal working of the mechanisms, as well as showing me what it takes for a highly efficient and motivated team to function together. Everyday is a new challenge and nothing is more enjoyable than being able to come up with solutions in a quick and effective manner. In my journey so far, I have come to know amazing individuals in my team. Through hard work, dedication, a deep commitment to excellence and a great team that I am blessed with, I am confident that we can achieve great things together. Hoping to see much more action in the coming days.

PROJECT MANAGER

Anmol Sah



Hi I'm Anmol Sah, and Ashwa is like a second home to me. Not only has being a part of Ashwa expanded my academic horizons, but it has also improved my interpersonal skills in other areas. As a car aficionado, I had many ideas regarding the appearance and functionality of the automobile, but ashwa played a crucial role in turning my thoughts into working prototypes. I've witnessed first-hand as a part of this team how much effort, cooperation, and commitment it takes to create just one prototype. I'm grateful to have such an amazing team that is always encouraging and motivating. Being at Ashwa is the most unforgettable part of my experience at this college.

CHIEF ENGINEER

Darren Nishan Patrao



Hey! I'm Darren, the Chief Engineer (FS) for the 2024 batch. I am a huge motorsports fan and therefore joining a club that pursues mobility solutions and builds formula student cars was a no brainer for me. I don't shy away from getting my hands dirty, having spent countless hours in the workshop working on the car, exploring and learning the journey until now has certainly made me a better engineer. My responsibilities at ashwa include designing the chassis and overseeing all technical decisions regarding the car, I am also incharge of workshop management. Being part of a team has also helped me understand team dynamics and develop managerial and soft skills. My team has grown and progressed over time and we are motivated now more than ever to build something fantastic. I'm currently working with my team to build the FS Combustion and the FS Electric car for the 2024 season.

ALUMNUS FOCUS

A SERIES : Meet our
Alumnus and what they
have to say

TARUN KASA

Technology Risk Consultant at KPMG India

Sponsorship, Marketing and PR Lead | 2019 batch



Introduction

During my first semester at RVCE, I became a member of Ashwa Racing. As a part of the Sponsorship, Marketing, and Public Relations subsystem in Ashwa, the four-year journey was a valuable and fulfilling experience. Currently, I am employed in the advisory practice at KPMG India.

What motivated you to be a part of Ashwa Racing?

Joining Ashwa Racing was motivated by my love for sports cars and Ferrari.

What was your experience like? What are some of the highlights of your time at Ashwa?

The knowledge gained from my time with Ashwa has been vital in my ability to overcome real-world obstacles. I have countless cherished memories from my experience.

Key highlight – Finishing 2nd at Formula Hybrid 2018 after an 42 Lap run during the endurance event in New Hampshire Motor Speedway, USA.

"When you work with a team, sometimes you don't just learn lessons for a project, but for life."

TARUN KASA

HUMANS OF
FORMULA BHARAT



What advise would you like to give to the current team and students in engineering?

The absence of teamwork yields no outcomes, and no task should be considered trivial. Every task presents an opportunity to learn and grow.

SPONSOR FOCUS

An introduction to the companies
and industries that have been
pillars of support!



Meritor, Inc. is a company whose expertise lies in the manufacture of automobile parts and components. A large part of their business includes axles, brake and safety systems, drivelines, suspensions, trailers, and aftermarket products for defense industries and commercial vehicles including truck, trailer, bus/coach, and off-highway. Meritor also engineers and has long standing partnerships with OEMs like Volvo, Navistar and Daimler.

Infineon sells chip cards and security devices in addition to semiconductors and systems for the automotive, industrial, and multimarket segments. Infineon offers semiconductor devices for use in safety systems, comfort electronics (such as steering, shock absorbers, and air conditioning), and powertrains (engine and transmission control) (ABS, airbags, ESP). Sensors, power semiconductors, and microcontrollers are all part of the product line. They have established a reputation for excellence in the field. Ashwa expresses gratitude for their support!



Log9 Materials are building solutions towards 'Pioneering Responsible Energy'. They are paving a new path by harnessing the power of science for positive change. The innovative, deep tech-driven approach ensures the creation having a significant impact in the future inculcating the habit of "being responsible" at every step by the work done at each step. Ashwa Mobility is grateful to have LOG9 Materials on board for the current race season!



Known to be the largest brake system supplier in India, Brakes India Pvt. Ltd. has over 50 years of experience behind it. Brakes India Private Limited is divided into three primary business units: brakes, foundries, and polymers. The extensive product line includes Dry and Wet multiple plate disc brakes for agricultural tractors, S-cam, Hydraulic drum brakes, Disk brakes, and Electromagnetic Retarders for commercial vehicles, as well as calipers, actuation, drum brakes, valves, hose ABS, and brake fluid for passenger vehicles. They are all set to make their mark on the world platform. Ashwa thanked Brakes India for their continued support to the team.

Texas Instruments Incorporated is an American technological business. It creates and produces integrated circuits and semiconductors, which it then sells to producers and designers of electronics around the world. According to sales volume, it is one of the top 10 semiconductor businesses in the world. Calculators, microcontrollers, and multi-core processors are among the items that TI also makes using its TI digital light processing technology and education technologies. As of 2016, the business had 45,000 patents worldwide. Thanks to TI for their advice and assistance, Ashwa.



Pegasys Systems Pvt Ltd is a company that offers expertise in designing and building PC and PLC controlled special purpose machines. The company caters to the needs of various industries in the field of automobile, electrical switchgear, refrigeration and air conditioning, pharmaceutical and other engineering industries. The company's products are well known for their durability, high efficiency and superior quality. Ashwa Racing is indebted to Pegasys systems for their never ending support towards the team.

Germany's Gerlingen is home to the international engineering and technology corporation Bosch. The four business segments of mobility (hardware and software), consumer products (including home appliances and power tools), industrial technology (including drive and control), and energy and building technology make up Bosch's key operational areas. The Ashwa team thanks Bosch for their support and encouragement.



Littelfuse, Inc is an American electronic manufacturing company. The company operates between three business unit segments: Electronics, Industrial, and Automotive. They primarily produce circuit protection products but also manufactures a variety of electronic switches and automotive sensors and is considered an industry leader. The support that Ashwa has received from Littlefuse has been incredible and we'd like to thank them for it.

Fie Tech is an engineering organization engaged in design, fabrication and commissioning of turnkey fiber glass projects in the field of architectural, interiors, landscaping and chemical industries. The company's innovative products range from aesthetic architectural items to rugged marine equipment. FIE-TECH manufactures a wide range of products from fiber reinforced plastic (FRP), custom made products for specific requirements. The company's specialty lies in the in-house design and development of all the products. Ashwa Racing thanks Fie Tech for their continued support towards the team and project.



Since their establishment in 1997, Magod Laser has steadily increased their reputation for competence by assisting clients in implementing laser-based manufacturing processes and applications across a range of industries. Magod has been a tremendous help to the Ashwa team, and we will always be appreciative.



Reflecton are the leading providers of auto detailing services for motorcycle and car enthusiasts and want to take auto detailing in India to a whole new level. Their goal is to use cutting-edge techniques while delivering excellent results on every project and providing smart surface solutions for any surface that needs to be treated or applied to an automobile component. Their key focus is based on quality, attention to detail and delivery. They also undertake on-demand services such as: motorcycle restoration, car and motorcycle spray painting, performance enhancement, wraps and peel-able paints, and even trust preventative treatments. We are ever grateful for Reflecton's contributions.



Divya Fuel Point has always been a step ahead of its competitors, whether it's through the introduction of automated fueling stations known as e-fuel stations in 2008 or the introduction of new retail outlet formats in 2006 that are specifically created to meet the needs of different customer segments. This transformational path has been not only thrilling but also extremely rewarding. At HPCL Retail, they believe in the maxim: Think big, dream big and achieve big. Thank you for your continued support to Ashwa Mobility!



From our Foundation to the date, Ashwa's magnificent partners!

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From our Foundation to the date, Ashwa's magnificent partners!

2003



2023

TEAM DIRECTORY

The team members of the
2024 batch

FACULTY ADVISOR

Dr. Ravindra S. Kulkarni

FOUNDER & COACH

Arunraj Subbaraj

DRIVETRAIN & HR

Anmol Sah	9589299888	anmol.ashwa@gmail.com
Vikyathraj Shetty K	9148786777	vikhyathraj.ashwa@gmail.com
Dennis Mathew	8197950390	Dennis.ashwa@gmail.com
Shivaraam	9489294813	shivaraam.ashwa@gmail.com
Ritik Raj	7033589289	ritikraj.ashwa@gmail.com
Om Ravi	8830690984	om.ashwa@gmail.com
Aman Kapoor	7091580376	amankapoor.ashwa@gmail.com

ELECTRICAL & TESTING

Nihar HG	8951231601	niharhg.ashwa@gmail.com
Sai Nithin Yadamreddy	7674955268	sainithin.ashwa@gmail.com
Saathvik BS	8792897224	saathvikbs.ashwa@gmail.com
Nipun Sagiraju	8310795066	nipunsagiraju.ashwa@gmail.com
Alaska Tengli	9886094514	alaska.ashwa@gmail.com
Shriram J Sharma	8095654796	shriram.ashwa@gmail.com
Dhanush Prabhakar	8861403498	dhanushprabhakar.ashwa@gmail.com
Neeleshadki	7795950221	neelesh.ashwa@gmail.com
Yashaswini MR	8105710945	yashaswinimr.ashwa@gmail.com
Hima Priya KN	7619456663	himapriyakn.ashwa@gmail.com
MV Sumukh	8971126596	mvsumukh.ashwa@gmail.com
Mir Ibrahim Faiz	7026256877	miribrahimfaiz.ashwa@gmail.com
Chirag C Rao	8277019525	chiragcrao.ashwa@gmail.com
R Vishnu	9538745573	rvishnu.ashwa@gmail.com

STEERING & FINANCE

Manav Purohit	9606723557	manavpurohit.ashwa@gmail.com
Saikumar A Patil	8951552471	saikumarapatil.ashwa@gmail.com

CHASSIS AND WORKSHOP

Darren Nishan Patrao	9035263349	darren.ashwa@gmail.com
Mahendra Singh Rawat	7760898790	mahendrasr.ashwa@gmail.com
Chethan P	6360059199	chethanp.ashwa@gmail.com
Suryaansh Agrawal	7618768699	suryaansh.ashwa@gmail.com
Aditya B	8660570429	adityab.ashwa@gmail.com
Lepaxi Bonageri	8296288067	lepaxi.ashwa@gmail.com
Harsh Goyal	6260022507	harshgoyal06.ashwa@gmail.com

ENGINE

Surag R	9845000387	surag.ashwa@gmail.com
Tejonidhi G.K.	9632597099	tejonidhi.ashwa@gmail.com
Dennis Mathew	8197950390	Dennis.ashwa@gmail.com
Rakesh Shankar	6364179911	rakeshshankar.ashwa@gmail.com

BRAKES AND LOGISTICS

Aman Kapoor	7091580376	amankapoor.ashwa@gmail.com
Kenny Nathan Lokku	7022664120	kenny.ashwa@gmail.com
Hitaish PV	9066556777	hitaishpv.ashwa@gmail.com
Manikant	7975914297	manikant.ashwa@gmail.com
Vijay Kumar P R	8088165516	vijaykumarpr.ashwa@gmail.com
Devansh Nandi	8879274336	devanshd.ashwa@gmail.com

SUSPENSION AND ADMIN

Anshuneel Dumbhare	7038609606	anshuneel.ashwa@gmail.com
D Gnaneshwar Raju	9945744228	gnaneshwar.ashwa@gmail.com
Utkarsh Sagar	8678085772	utkarshsagar.ashwa@gmail.com
Adithya Thantri	9972872862	adithyathantri.ashwa@gmail.com

INFORMATION TECHNOLOGY

Shravan Shenoy
Anish AS

7349481830
6363652078

shravanshenoy.ashwa@gmail.com
anish.ashwa@gmail.com

SPONSORSHIP, MARKETING, AND PR

Akula Yashas Anand
Meghana C S
Antriksh Jain
Pratyush Joshi
Anusha Khot
Aadhya Ravindra
Kabir Shukla
Ashwin Rajhan

7259696797
7022591438
8105818528
7451873446
9353705093
9986189900
8506049900
9945966431

yashas.ashwa@gmail.com
meghanacs.ashwa@gmail.com
antrikshjain.ashwa@gmail.com
pratyushjoshi.ashwa@gmail.com
anusha.ashwa@gmail.com
aadhya.ashwa@gmail.com
kabirs.ashwa@gmail.com
ashwinrajhan.ashwa@gmail.com



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arf.socialnet@gmail.com



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[@ashwaracing](https://www.twitter.com/ashwaracing)

बड़हवा MOBILITY FOUNDATION



THANK YOU ONE AND ALL

LIVE IT. LOVE IT. RACE IT.